

Genetic report for longevity

Hereditary Genetic Diseases

ClinVar screening of 81 actionable genes in the list ACMG SF v3.2 (hereditary cancer, cardiomyopathies and arrhythmias, familial hypercholesterolemia, aortopathies and others) and of 39 monogenic genes out of the list, with real chip coverage and imputation

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This report seeks variants of hereditary diseases with possible medical action: predisposition to cancer (BRCA1/2, Lynch, APC, TP53, PALB2 and others), cardiomyopathies and arrhythmias, familial hypercholesterolemia (LDLR, APOB, PCSK9), aortopathies (FBN1, TGFBR1/2, COL3A1), hemochromatosis and others, following the 81 genes of the list ACMG SF v3.2 of secondary findings, plus 39 relevant genes (CHEK2, ATM, CDH1, CDKN2A, MODY, Parkinson, dementia and thrombophilia). All variants P/LP of ClinVar in these genes were crossed with the chip (SNVs and deletion/insertion probes) and with imputation (only high confidence). The focus is on dominant and actionable diseases; the states of carrier of recessive diseases are in the State of Portador report. The chips have many false positives in rare variants (in UK Biobank, only about 16% of the very rare variants read by chip were confirmed by sequencing): any positive finding requires clinical confirmation before any decision.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

Contents

• Summary	3
• Screening and coverage results	5
• Hereditary cancer.....	11
• Heart, aorta and family cholesterol.....	13
• Other actionable genes and out of the list	14
• Mechanisms of ageing.....	15
• Practical part	16
• Limitations.....	17
• Methods and sources	18

Summary

20

variants

20

in the chip

0

imputed
(DR2 $\geq$ 0.8)

0

not found

0

Scores with
percentile

Overall reading. No pathogenic or probable pathogenic variant was found in 81 genes of the list ACMG SF v3.2 or in 39 extra monogenic genes. The screening crossed all variants P/LP of ClinVar in these genes (58 123 in ACMG genes) with the chip and imputation; 12 239 (21%), including 8 643 of 24 899 with ≥ 2 stars (35%). The founder and moderate penetrance variants are all absent: BRCA1 185delAG and 5382insC, CHEK2 I157T, APC I1307K, HOXB13 G84E, MITF E318K, APOB R3527Q, TTR V122I and V30M and LRRK2 G2019S. PCSK9 R46L is also absent (it is the protective variant; without judgment). Nine readings with the letter of a P/LP allele are artifacts and do not count: six are probes of multiple variants or deletions/insertions read in the position of a SNV (APC, ATM, TSC2, LMNA, KCNH2, RYR1), one is the homozygosity of the NF1 of the profile screening (gene with pseudogenes) and two are indel probes of the BRCA1 and the KCNH2 whose "heterozygosity" is proven, a common benign variant in the same position (BRCA1 P871L and KCNH2 K897T, confirmed by imputation). It is a good example of the main limit of chips: false positives in rare variants are frequent. And the opposite is also true: this negative result does not exclude a pathogenic variant, because almost 80% of known variants, large deletions and new variants are not read. The recessive carrier states (SERPINA1 PiMZ, HFE H63D) are in the Carrier Status report.

Favourable

- **BRCA1 (185delAG) rs80357914 AA:** BRCA1 185delAG (c.68_69del): founder variant; chip probe itself.
- **BRCA1 (5382insC) rs80357906 TT:** BRCA1 5382insC (c.5266dup): founder variant; own probe of the chip.
- **CHEK2 (I157T) rs17879961 AA:** CHEK2 I157T: low penetrance variant (breast cancer, prostate and colon ~ 1.2 – $1.5\times$); classification in the ClinVar with conflicts, so outside the P/LP count.
- **APC (I1307K) rs1801155 TT:** APC I1307K: low penetrance risk factor for colon cancer (~ 1.5 – $2\times$).
- **HOXB13 (G84E) rs138213197 CC:** HOXB13 G84E: risk of prostate cancer ~ 3 – $4\times$ (early).
- **MITF (E318K) rs149617956 GG:** MITF E318K: melanoma and kidney cancer ~ 2 – $3\times$.
- **APOB (R3527Q) rs5742904 CC:** APOB R3527Q: family defect of ApoB (high LDL since childhood).
- **TTR (V122I) rs76992529 GG:** TTR V122I: hereditary cardiac amyloidosis.
- **TTR (V30M) rs28933979 GG:** TTR V30M: hereditary amyloidosis by transthyretin with nerve damage ("peet disease").
- **LRRK2 (G2019S) rs34637584 GG:** LRRK2 G2019S: the most frequent genetic cause of Parkinson's disease (penetration ~ 25 – 40% until age 80).

Unfavourable

None.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 20 chip genotypes, 1 needed complement, 1 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 6 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect

alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Screening and coverage results

Coverage in 81 genes ACMG SF v3.2: ClinVar (version of 28-09-2026) has 58 123 pathogenic variants (P/LP, without conflicts) in these genes, 24 899 of them with ≥ 2 review stars. The chip and the imputation read 12 239 (21%): 10 465 by SNV probes, 1 773 by probes of small deletions/insertions and only 1 by imputation (the rare variants almost never pass the confidence threshold $DR2 \geq 0.8$). Of the variants with ≥ 2 stars, 8 643 (35%) are read. In 39 genes out of the list: 674 of 14 506 (5%). Large deletions and duplications are always left out (e.g., BRCA1, MSH2/EPCAM and LDLR), most of the private variants of each family and the genes with pseudogenes or repeated regions (PMS2, TTN in part). A negative result does not exclude a pathogenic variant: for this, clinical sequencing is required (gene panel or complete genome to 30x).

P/LP plausible findings in 120 genes: none. There were 9 readings with the letter of the P/LP allele; all are explicable artifacts (following sections): probes of multiple variants or deletions/insertions read as if they were SNVs, and two indel probes whose "heterozygosity" is, in fact, a benign common variant in the same position. No P/LP variant was read by high-confidence imputation.

Cover table: 81 genes ACMG SF v3.2

Gene	Group	Variants P/ LP in ClinVar	With ≥ 2 stars	Reads: chip (SNV)	Reads: indel probe	Read only by imputation	Total read (%)	Dealing with ≥ 2 stars (%)	Plausible findings
APC	Hereditary cancer	2 595	1 113	315	52	0	367 (14%)	286 (26%)	0
BMPRI1A	Hereditary cancer	304	100	51	6	0	57 (19%)	30 (30%)	0
BRCA1	Hereditary cancer	4 116	2 869	721	413	0	1 134 (28%)	1 025 (36%)	0
BRCA2	Hereditary cancer	5 675	3 714	874	410	0	1 284 (23%)	1 161 (31%)	0
MAX	Hereditary cancer	53	13	4	0	0	4 (8%)	3 (23%)	0
MEN1	Hereditary cancer	680	265	143	29	0	172 (25%)	109 (41%)	0
MLH1	Hereditary cancer	1 894	1 061	422	102	0	524 (28%)	464 (44%)	0
MSH2	Hereditary cancer	2 093	1 100	392	109	0	501 (24%)	444 (40%)	0
MSH6	Hereditary cancer	2 293	1 132	206	98	0	304 (13%)	269 (24%)	0
MUTYH	Hereditary cancer	419	192	89	3	0	92 (22%)	72 (38%)	0
NF2	Hereditary cancer	296	61	42	1	0	43 (15%)	22 (36%)	0
PALB2	Hereditary cancer	1 453	796	77	20	0	97 (7%)	88 (11%)	0
PMS2	Hereditary cancer	935	482	159	18	0	177 (19%)	148 (31%)	0
PTEN	Hereditary cancer	1 330	569	290	39	0	329 (25%)	268 (47%)	0
RB1	Hereditary cancer	977	294	166	13	0	179 (18%)	97 (33%)	0
RET	Hereditary cancer	197	68	54	5	0	59 (30%)	44 (65%)	0
SDHAF2	Hereditary cancer	50	25	2	1	0	3 (6%)	3 (12%)	0
SDHB	Hereditary cancer	287	153	84	10	0	94 (33%)	75 (49%)	0
SDHC	Hereditary cancer	101	48	17	1	0		17 (35%)	0

							18 (18%)		
SDHD	Hereditary cancer	158	78	38	2	0	40 (25%)	33 (42%)	0
SMAD4	Hereditary cancer	307	95	64	17	0	81 (26%)	43 (45%)	0
STK11	Hereditary cancer	402	116	94	13	0	107 (27%)	60 (52%)	0
TMEM127	Hereditary cancer	125	46	1	0	0	1 (1%)	1 (2%)	0
TP53	Hereditary cancer	1 154	474	288	16	0	304 (26%)	252 (53%)	0
TSC1	Hereditary cancer	768	200	96	13	0	109 (14%)	64 (32%)	0
TSC2	Hereditary cancer	1 477	395	251	27	0	278 (19%)	153 (39%)	0
VHL	Hereditary cancer	435	189	145	17	0	162 (37%)	113 (60%)	0
WT1	Hereditary cancer	175	26	31	0	0	31 (18%)	13 (50%)	0
ACTA2	Heart and aorta	44	20	20	0	0	20 (45%)	14 (70%)	0
ACTC1	Heart and aorta	27	7	7	0	0	7 (26%)	4 (57%)	0
APOB	Heart and aorta	258	50	23	1	0	24 (9%)	11 (22%)	0
BAG3	Heart and aorta	161	49	10	0	0	10 (6%)	8 (16%)	0
CALM1	Heart and aorta	21	4	1	0	0	1 (5%)	1 (25%)	0
CALM2	Heart and aorta	21	4	0	0	0	0 (0%)	0 (0%)	0
CALM3	Heart and aorta	9	3	0	0	0	0 (0%)	0 (0%)	0
CASQ2	Heart and aorta	80	34	6	2	0	8 (10%)	4 (12%)	0
COL3A1	Heart and aorta	963	215	475	12	0	487 (51%)	110 (51%)	0
DES	Heart and aorta	131	42	27	0	0	27 (21%)	19 (45%)	0
DSC2	Heart and aorta	107	21	6	0	0	6 (6%)	3 (14%)	0
DSG2	Heart and aorta	137	34	13	0	0	13 (9%)	9 (26%)	0
DSP	Heart and aorta	845	264	92	16	0	108 (13%)	71 (27%)	0
FBN1	Heart and aorta	3 684	1 051	1 131	38	0	1 169 (32%)	570 (54%)	0
FLNC	Heart and aorta	654	126	1	0	0	1 (0%)	1 (1%)	0
KCNH2	Heart and aorta	768	262	175	18	0	193 (25%)	112 (43%)	0
KCNQ1	Heart and aorta	569	281	201	13	0	214 (38%)	153 (54%)	0
LDLR	Heart and aorta	1 973	1 092	856	90	0	946 (48%)	707 (65%)	0
LMNA		556	186	172	12	0		105 (56%)	0

	Heart and aorta						184 (33%)		
MYBPC3	Heart and aorta	1 083	504	224	51	0	275 (25%)	216 (43%)	0
MYH11	Heart and aorta	78	6	7	0	0	7 (9%)	2 (33%)	0
MYH7	Heart and aorta	440	200	244	1	0	245 (56%)	159 (80%)	0
MYL2	Heart and aorta	18	7	9	0	0	9 (50%)	5 (71%)	0
MYL3	Heart and aorta	5	3	3	0	0	3 (60%)	3 (100%)	0
PCSK9	Heart and aorta	17	4	12	0	0	12 (71%)	4 (100%)	0
PKP2	Heart and aorta	361	156	68	10	0	78 (22%)	53 (34%)	0
PRKAG2	Heart and aorta	27	7	14	1	0	15 (56%)	7 (100%)	0
RBM20	Heart and aorta	110	11	7	0	0	7 (6%)	7 (64%)	0
RYR2	Heart and aorta	263	64	102	0	0	102 (39%)	50 (78%)	0
SCN5A	Heart and aorta	567	195	133	10	0	143 (25%)	83 (43%)	0
SMAD3	Heart and aorta	212	72	30	5	0	35 (17%)	22 (31%)	0
TGFBR1	Heart and aorta	112	23	31	0	0	31 (28%)	10 (43%)	0
TGFBR2	Heart and aorta	126	39	55	1	0	56 (44%)	25 (64%)	0
TMEM43	Heart and aorta	3	1	1	0	0	1 (33%)	1 (100%)	0
TNNC1	Heart and aorta	11	0	2	0	0	2 (18%)	0 (0%)	0
TNNI3	Heart and aorta	80	25	44	0	0	44 (55%)	22 (88%)	0
TNNT2	Heart and aorta	67	29	39	1	0	40 (60%)	22 (76%)	0
TPM1	Heart and aorta	53	13	25	0	0	25 (47%)	11 (85%)	0
TRDN	Heart and aorta	57	19	1	0	0	1 (2%)	1 (5%)	0
TTN	Heart and aorta	6 062	1 347	128	24	0	152 (3%)	120 (9%)	0
BTD	Metabolism	251	110	23	1	0	24 (10%)	22 (20%)	0
GAA	Metabolism	856	481	26	1	0	27 (3%)	26 (5%)	0
GLA	Metabolism	1 173	417	198	4	0	202 (17%)	154 (37%)	0
OTC	Metabolism	446	66	306	5	0	311 (70%)	54 (82%)	0
ACVRL1	Others	571	210	28	1	0	29 (5%)	21 (10%)	0
ATP7B	Others	841	391	127	12	0	139 (17%)	113 (29%)	0
CACNA1S	Others	130	22	10	0	0	10 (8%)	6 (27%)	0
ENG	Others	626	242	25	1	0	26 (4%)	22 (9%)	0

HFE	Others	47	13	2	0	0	2 (4%)	1 (8%)	0
HNFB1A	Others	402	274	15	3	0	18 (4%)	17 (6%)	0
RPE65	Others	341	214	12	0	1	13 (4%)	13 (6%)	0
RYR1	Others	798	237	159	3	0	162 (20%)	89 (38%)	0
TTR	Others	132	78	23	1	0	24 (18%)	23 (29%)	0

Cover table: 39 genes out of list

Gene	Group	Variants P/ LP in ClinVar	With ≥2 stars	Reads: chip (SNV)	Reads: indel probe	Read only by imputation	Total read (%)	Dealing with ≥2 stars (%)	Plausible findings
ATM	Cancer (outside ACMG)	3 821	1 800	201	41	0	242 (6%)	212 (12%)	0
BARD1	Cancer (outside ACMG)	627	305	16	4	0	20 (3%)	19 (6%)	0
BRIP1	Cancer (outside ACMG)	887	454	34	12	0	46 (5%)	44 (10%)	0
CDH1	Cancer (outside ACMG)	634	279	23	3	0	26 (4%)	24 (9%)	0
CDKN2A	Cancer (outside ACMG)	168	87	23	4	0	27 (16%)	26 (30%)	0
CHEK2	Cancer (outside ACMG)	1 045	447	55	8	1	64 (6%)	60 (13%)	0
EPCAM	Cancer (outside ACMG)	50	10	4	0	0	4 (8%)	2 (20%)	0
HOXB13	Cancer (outside ACMG)	2	1	1	0	0	1 (50%)	1 (100%)	0
MITF	Cancer (outside ACMG)	145	28	3	0	0	3 (2%)	2 (7%)	0
NTHL1	Cancer (outside ACMG)	156	77	1	0	0	1 (1%)	1 (1%)	0
POLD1	Cancer (outside ACMG)	8	4	1	0	0	1 (12%)	1 (25%)	0
POLE	Cancer (outside ACMG)	459	10	1	0	0	1 (0%)	1 (10%)	0
RAD51C	Cancer (outside ACMG)	330	172	30	2	0	32 (10%)	28 (16%)	0
RAD51D	Cancer (outside ACMG)	249	118	9	4	0	13 (5%)	12 (10%)	0
ABCG5	Others (outside ACMG)	77	21	1	0	0	1 (1%)	1 (5%)	0
ABCG8	Others (outside ACMG)	62	19	4	0	0	4 (6%)	4 (21%)	0
APOA5		17	2	0	0	0	0 (0%)	0 (0%)	0

	Others (outside ACMG)								
APP	Others (outside ACMG)	28	10	2	0	0	2 (7%)	1 (10%)	0
F2	Others (outside ACMG)	41	7	5	0	0	5 (12%)	3 (43%)	0
F5	Others (outside ACMG)	114	21	2	0	0	2 (2%)	1 (5%)	0
GCK	Others (outside ACMG)	704	507	83	0	0	83 (12%)	74 (15%)	0
GRN	Others (outside ACMG)	137	36	1	0	0	1 (1%)	1 (3%)	0
HNF1B	Others (outside ACMG)	293	80	0	0	0	0 (0%)	0 (0%)	0
HNF4A	Others (outside ACMG)	168	112	7	1	0	8 (5%)	8 (7%)	0
KCNE1	Others (outside ACMG)	21	5	1	0	0	1 (5%)	0 (0%)	0
KCNJ2	Others (outside ACMG)	65	22	22	0	0	22 (34%)	13 (59%)	0
LIPA	Others (outside ACMG)	160	54	6	1	0	7 (4%)	6 (11%)	0
LPL	Others (outside ACMG)	166	52	9	0	0	9 (5%)	5 (10%)	0
LRRK2	Others (outside ACMG)	11	5	2	0	0	2 (18%)	1 (20%)	0
MAPT	Others (outside ACMG)	36	14	2	0	0	2 (6%)	1 (7%)	0
NOTCH3	Others (outside ACMG)	329	124	6	0	0	6 (2%)	3 (2%)	0
PKD1	Others (outside ACMG)	2 456	601	2	0	0	2 (0%)	2 (0%)	0
PKD2	Others (outside ACMG)	459	132	4	0	0	4 (1%)	1 (1%)	0
PROC	Others (outside ACMG)	113	40	5	0	0	5 (4%)	5 (12%)	0
PROS1	Others (outside ACMG)	147	24	2	1	0	3 (2%)	2 (8%)	0
PSEN1	Others (outside ACMG)	152	51	8	0	0	8 (5%)	6 (12%)	0
PSEN2	Others (outside ACMG)	10	2	6	0	0	6 (60%)	2 (100%)	0

SERPINC1	Others (outside ACMG)	154	56	10	0	0	10 (6%)	9 (16%)	0
SNCA	Others (outside ACMG)	5	2	0	0	0	0 (0%)	0 (0%)	0

Ageing mechanisms: Genomic instability; DNA Damage and DNA Repair Deficit

Favourable

None.

Unfavourable

None.

Hereditary cancer

BRCA1/BRCA2 (mamma, ovary, prostate, pancreas), Lynch syndrome (MLH1, MSH2, MSH6, PMS2, EPCAM; colon and others), APC and MUTYH (polyposis), TP53 (Li-Fraumeni), PALB2, CHEK2, ATM and others. In addition to the complete screening, we read here the most well-known founder and moderate penetration variants that the chip has: BRCA1 185delAG and 5382insC, CHEK2 I157T, APC I1307K, HOXB13 G84E and MITF E318K. Penetration of a pathogenic variant: BRCA1 ~60–70% of breast cancer in women and ~1–2% in men with more prostate and pancreas cancer; BRCA2 ~5–8% of breast cancer in men and about 2× prostate cancer; Lynch 30–60% colon cancer.

Ageing mechanisms: DNA Damage and DNA Repair Deficit; Genomic instability; Somatic Mutation Accumulation

Favourable

- BRCA1 (185delAG) rs80357914 AA
- BRCA1 (5382insC) rs80357906 TT
- CHEK2 (I157T) rs17879961 AA
- APC (I1307K) rs1801155 TT
- HOXB13 (G84E) rs138213197 CC
- MITF (E318K) rs149617956 GG

Unfavourable

None.

Favourable variants (6)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
BRCA1 (185delAG)	rs80357914 17:43124028 · CTCT/CT/CTCTCT	AA	chip	deletion (0; —)	strong	BRCA1 185delAG (c.68_69del): founder variant; chip probe itself. Homozygous deletion probe (AA): a carrier would show two different letters. Without the variant.
BRCA1 (5382insC)	rs80357906 17:43057063 · GGG/GGGG	TT	chip	insertion (0; —)	strong	BRCA1 5382insC (c.5266dup): founder variant; own probe of the chip. Homozygous insertion probe (TT): a carrier would show two different letters. Without the chip's own probe (rs80357906)
CHEK2 (I157T)	rs17879961 22:28725099 · A/C/G/T	AA	chip	G (0; 0%)	moderate	CHEK2 I157T: low penetrance variant (breast cancer, prostate and colon ~1.2–1.5×); classification in the ClinVar with conflicts, so outside the P/LP count.
APC (I1307K)	rs1801155 5:112839514 · T/A	TT	chip	A (0; —)	moderate	APC I1307K: low penetrance risk factor for colon cancer (~1.5–2×).
HOXB13 (G84E)	rs138213197 17:48728343 · C/T	CC	chip	T (0; 0%)	strong	HOXB13 G84E: risk of prostate cancer ~3–4× (early).
MITF (E318K)	rs149617956 3:69964940 · G/A/T	GG	chip	A (0; 0%)	moderate	MITF E318K: melanoma and kidney cancer ~2–3×.

Non-judgmental variants (informative) (4)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
BRCA1 (rs397508986 probe)	rs397508986 17:43092919 · G/AA	AG	chip	—	strong	Chip probe for a rare pathogenic variant of BRCA1 (G>AA, 3 stars) reads as "GA". Artifact: in the same position there is a benign variant BRCA1 P871L (rs799917), and imputation confirms that this is heterozygosity (G/A, DR2=1.00). It is

						not a carrier of the pathogenic variant.
APC (multiple variant probe)	rs1580623270 5:112838101 · C/A/T	AA	chip (probe of another variant in the same position)	—	strong	Position of a nonsense variant of APC; the probe of the chip is a multiple variant (MNV) in the same position and the letters do not match. Homozygosity for a polyposis variant would be incompatible with the absence of disease. Probe coding artifact. Artifact: does not count as finding.
ATM (deletion/insertion probe)	rs780905851 11:108345889 · T/A/C/G	GG	chip (probe of another variant in the same position)	—	strong	Position of a pathogenic variant of ATM; the probe of the chip is another variant (deletion/insertion TG>AA, rs587781353) in the same position. Homozygosity would give ataxia-telangiectasia. Artifact. Artifact: does not count as finding.
TSC2 (multiple variant probe)	rs45517205 16:2071508 · A/C/G/T	CC	chip (probe of another variant in the same position)	—	strong	Position of a splicing variant of TSC2 (tuberous sclerosis); MNV probe with letters that do not match. Artifact. Artifact: does not count as finding.

Heart, aorta and family cholesterol

Cardiac disorders (MYBPC3, MYH7, TNNT2, LMNA, TTN, FLNC, DSP, PKP2), arrhythmias (KCNQ1, KCNH2, SCN5A, RYR2, CALM1–3), aortopathies (FBN1, TGFB1/2, SMAD3, ACTA2, COL3A1, MYH11) and familial hypercholesterolemia (LDLR, APOB, PCSK9). Variants read in prominence: APOB R3527Q (the most frequent variant of family defect of ApoB), PCSK9 R46L (protective variable of function loss) and TTR V122I and V30M (cardiac and hereditary neurological amyloidosis).

Ageing mechanisms: Vascular Aging; Arterial stiffness; Fragmentation of Elastin

Favourable

- APOB (R3527Q) rs5742904 CC
- TTR (V122I) rs76992529 GG
- TTR (V30M) rs28933979 GG

Unfavourable

None.

Favourable variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
APOB (R3527Q)	rs5742904 2:21006288 · C/A/T	CC	chip	T (0; —)	strong	APOB R3527Q: family defect of ApoB (high LDL since childhood).
TTR (V122I)	rs76992529 18:31598655 · G/A	GG	chip	A (0; 0%)	strong	TTR V122I: hereditary cardiac amyloidosis.
TTR (V30M)	rs28933979 18:31592974 · G/A/C	GG	chip	A (0; —)	strong	TTR V30M: hereditary amyloidosis by transthyretin with nerve damage ("peet disease").

Non-judgmental variants (informative) (4)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
PCSK9 (R46L)	rs11591147 1:55039974 · G/A/T	GG	chip	—	strong	PCSK9 R46L: the T allele is of loss of function and protector (LDL ~15% lower, less coronary disease). G/G: without the protective variant.
LMNA (deletion/insertion probe)	rs2102817930 1:156115082 · A/G	GG	chip (probe of another variant in the same position)	—	strong	Position of a likely pathogenic variant of LMNA; the chip probe is a deletion/insertion (rs1064793882) in the same position. Artifact. Artifact: does not count as finding.
KCNH2 (deletion/insertion probe)	rs1137617 7:150951110 · A/C/G/T	GG	chip (probe of another variant in the same position)	—	strong	Position of a nonsense variant of KCNH2 (long QT); the probe is a deletion/insertion (rs794728490). In the same position there is a common benign variant. Artifact. Artifact: does not count as finding.
KCNH2 (rs794728448 probe)	rs794728448 7:150948445 · CT/G	GT	chip	—	strong	Probe of a rare pathogenic deletion/insertion of KCNH2 read as "TG". Artifact: in the next position is the common variant KCNH2 K897T (rs1805123), read as heterozygous in the chip and imputation (T/G); the probe is reading this variant.

Other actionable genes and out of the list

Malignant hyperthermia (RYR1, CACNA1S; anesthesia), hereditary hemorrhagic telangiectasia (ACVRL1, ENG), MODY (HNF1A, GCK, HNF4A), Fabry's disease (GLA), Pompe's disease (GAA), Parkinson's disease (LRRK2 G2019S, SNCA), family dementia (APP, PSEN1, PSEN2, MAPT, GRN), CADASIL (NOTCH3), polycystic kidney (PKD1/PKD2) and thrombophilia (F5, F2, SERPINC1, PROC, PROS1). Hemochromatosis (HFE), MUTYH, ATP7B, BTBD, GAA and RPE65 are recessive: the carrier state is in the carrier state report (HFE H63D heterozygous, without C282Y). Factor V Leiden and prothrombin are in the Advanced Cardiovascular Report (negative).

Ageing mechanisms: Protein aggregation; Lysosome dysfunction; Metabolic Homeostasis Decline

Favourable

- LRRK2 (G2019S) rs34637584 GG

Unfavourable

None.

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
LRRK2 (G2019S)	rs34637584 12:40340400 · G/A	GG	chip	A (0; 0%)	strong	LRRK2 G2019S: the most frequent genetic cause of Parkinson's disease (penetration ~25-40% until age 80).

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
RYR1 (multiple variant probe)	rs1599665128 19:38580437 · T/C	CC	chip (probe of another variant in the same position)	—	strong	Position of a variant of RYR1; MNV probe with letters that do not match. There is no evidence of susceptibility to malignant hyperthermia. Artifact. Artifact: does not count as finding.
NF1 (profile tracking)	rs137854557 17:31214524 · A/G/T	GG	chip	—	strong	NF1: homozygosity read in the ClinVar profile screening for a variant of neurofibromatosis type 1; it would be incompatible with life without disease, and NF1 has pseudogenes. Artifact. Artifact: does not count as finding.

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Somatic Mutation Accumulation	—	BRCA1 (185delAG) AA; BRCA1 (5382insC) TT; CHEK2 (I157T) AA; APC (I1307K) TT; HOXB13 (G84E) CC; MITF (E318K) GG	—
Protein Aggregation	—	LRRK2 (G2019S) GG	—
DNA Damage and DNA Repair Deficit	—	BRCA1 (185delAG) AA; BRCA1 (5382insC) TT; CHEK2 (I157T) AA; APC (I1307K) TT; HOXB13 (G84E) CC; MITF (E318K) GG	—
Metabolic Homeostasis Decline	—	LRRK2 (G2019S) GG	—
Lysosome dysfunction	—	LRRK2 (G2019S) GG	—
Vascular Aging	—	APOB (R3527Q) CC; TTR (V122I) GG; TTR (V30M) GG	—
Fragmentation of Elastin	—	APOB (R3527Q) CC; TTR (V122I) GG; TTR (V30M) GG	—
Genomic instability	—	BRCA1 (185delAG) AA; BRCA1 (5382insC) TT; CHEK2 (I157T) AA; APC (I1307K) TT; HOXB13 (G84E) CC; MITF (E318K) GG	—
Arterial stiffness	—	APOB (R3527Q) CC; TTR (V122I) GG; TTR (V30M) GG	—

Practical part

Ideas from screening, crossed with the genetic plan v2 and current medication. Everything is suggestion (not applied).

1. No action by genetic finding

Genetics: 0 P/LP plausible variants in 120 genes

There is no finding that justifies, by itself, changing screenings, exams or medication. Cancer screenings recommended for age (colon, skin, prostate when indicated) remain as for anyone.

V2 genetic plan and medication: No change in the plan. • *suggestion (not applied)*

2. Clinical sequencing if there is a family history

Genetics: 21% coverage of known P/LP variants; large unread deletions

Cancer in young age or several cases of the same type in the family, sudden death before 50 years, cardiomyopathy, aneurysm or dissection of the aorta, or very high cholesterol in relatives are reasons for a genetic consultation and a complete clinical panel or genome to 30. Without family history, it is an option, not a need.

V2 genetic plan and medication: No change in the plan. • *suggestion (not applied)*

3. Always confirm before deciding

Genetics: 9 artefact readings in this screening

If any test of consumption, chip or analysis of raw data shows a rare pathogenic variant, confirm in a clinical laboratory (with a new sample) before any decision.

V2 genetic plan and medication: No change in the plan. • *suggestion (not applied)*

4. Anesthesia: what is known

Genetics: RYR1/CACNA1S read only in part; Normal BCHE (Carrier Status Report)

There is no sign of susceptibility to malignant hyperthermia or pseudocholinesterase deficiency in the read variants, but the coverage of RYR1 is partial. If there is a family history of severe reactions to anesthesia, mention it before surgery.

V2 genetic plan and medication: No change in the plan. • *suggestion (not applied)*

Avoiding or with caution

- Medical decisions (preventive surgeries, changes in screening or medication) based on unconfirmed chip readings. *suggestion (not applied)*
- To take the negative result as a guarantee of absence of hereditary disease. *suggestion (not applied)*

Limitations

- **A negative result does not exclude** A hereditary disease: the chip and the imputation read about a fifth of the P/LP variants known in these genes and none of the variants not yet described. To exclude, it is necessary clinical sequencing (gene panel or complete genome 30×) with genetic counseling.
- **A positive result must be confirmed** by a clinical laboratory before any decision: for very rare variants, most positive readings of the chip are false positive.
- There are no large deletions and duplications, variants in repeated regions or with pseudogenes (PMS2, TTN parts), repeat expansions (e.g. myotonic dystrophy, Huntington) or deep intronic variants.
- The classification of ClinVar changes with time; the 28-09-2026 version was used, only with P/LP without conflicts. The variants of uncertain meaning were not reported.
- Only genes: does not include family history, which remains the best indicator to request a clinical test. It is not a diagnosis.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local imputation (Beagle 5.5 with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: .

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentis: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

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